Concept Note (optimized draft)

Plant-Fiber Materials for Plastic Substitution — SCUT × UNIDO IET/PST

Plant-Fiber Materials for Plastic Substitution: A Methodology and Conformity Partnership

1. Purpose

The purpose of this note is to propose a technical partnership between South China University of Technology (SCUT) and the Directorate of SDG Innovation and Economic Transformation (IET), and specifically its Division of Fair Production, Sustainability Standards and Trade (IET/PST).

The subject is narrow. A manufacturer that replaces plastic with a plant-fiber product has at present no accepted way of showing how much plastic it has displaced, and no accepted way of showing what the displacement costs in carbon and material terms. Both questions are answerable. Neither is answered in a consistent way today. We propose to answer them in a form that a buyer, a testing laboratory and a regulator can all use.

Three outputs are proposed over eighteen months: a draft methodology for measuring and reporting substitution, a mapping of the technologies that make substitution possible, and a technical dialogue among producing countries. None of the three requires UNIDO funding or creates a new institution. Each is designed to be usable inside the conformity services that IET/PST already provides to member states.

The division of work follows institutional competence. SCUT brings the laboratory, the production data and the testing platform. UNIDO brings what a university cannot supply on its own: the standing that makes a method credible to governments, and the channels that place it in front of the firms that need it.

What is requested at this stage is correspondingly modest, namely a technical focal point in IET/PST to review the scope set out in Section 6 and to advise on the form of engagement. No commitment is sought before that review has taken place.

2. The measurement problem

The technology is not the obstacle. Molded fiber containers, coated food-service paper, paper-based industrial packaging and fiber-based barrier materials are in commercial production, and in several product categories they compete with plastics on cost as well as on performance. What is missing is an agreed way of describing what they do.

Three technical questions account for most of the disagreement.

The first is biogenic carbon. A plant-fiber product stores carbon taken up during growth and releases it at end of life, which may be months or decades later. ISO 14067 establishes the principle but leaves the timing convention to the practitioner. Two studies of the same product can therefore differ by more than the difference between the product and the plastic it displaces.

The second is the substitution boundary. A claim of avoided emissions depends on assumptions about the displaced incumbent, about the end-of-life mix of recycling, composting and incineration, and about the functional unit. End-of-life assumptions alone can move a result by a wide margin, by a factor of two or more in our own test calculations. In practice those assumptions are chosen by the analyst rather than measured, and they are seldom disclosed in a comparable form.

The third is the end-of-life claim itself. “Recyclable” and “degradable” are governed by different instruments in different markets: EN 13432 for compostable packaging, ISO 14855 and ISO 17556 for laboratory degradation, national repulpability schemes for fiber recovery, and, in the European Union, the recyclability requirements of the Packaging and Packaging Waste Regulation (EU) 2025/40, which applies from 12 August 2026. A product can satisfy one regime and fail another, and the marketing claim rarely states which.

The consequence falls hardest on exporters in developing countries, for whom the substitution market is the accessible one. A complete third-party footprint study costs tens of thousands of euros, which a mid-sized converter cannot absorb for every product line. Without a cheaper and reproducible route to a defensible claim, such firms stay outside the market that substitution is meant to serve. That is a trade barrier in effect, whatever the intention behind it.

This is not a new diagnosis. UNIDO’s environmental footprint work has reached the same conclusion about comparability, as have the technical reviews behind the EU Product Environmental Footprint method (Commission Recommendation (EU) 2021/2279). The gap persists for a reason other than neglect: resolving these questions requires three capabilities in one place, namely control over the test method, access to production data from real lines, and enough independence from any single technology to be trusted by the others. Few institutions hold all three.

3. Alignment with the IET/PST mandate

IET’s mandate covers the development of new UNIDO services, new partnerships and new sources of finance. Within IET, PST works on the quality and conformity infrastructure that exporting firms in developing countries need in order to take part in sustainable trade: metrology, testing, certification, accreditation, and the standards that link them.

The outputs proposed here sit upstream of that work. They are the methodological layer that country-level conformity projects currently have to improvise, project by project, at the cost of comparability between projects. A validated method, published as a public good, lets those projects start from a common baseline instead of building one each time.

The model is not new to UNIDO. The “Clean Energy Innovation” University Alliance, launched at Shandong University with the support of ITPO Beijing, has shown that a university-hosted network can operate credibly under UNIDO’s convening umbrella while remaining academically independent. What is proposed here applies that model to plastics substitution and trade compliance.

Two limits should be stated plainly at the outset. UNIDO does not endorse technologies, and nothing in this proposal asks it to. Nor is the partnership intended to write standards. Standards are written in ISO/TC 6, ISO/TC 61 and CEN/TC 261, and we have no ambition to displace them. The output is a method at a lower level of formality, which can be offered to those committees as an input once it has been tested in use.

4. What SCUT would contribute

SCUT's contribution rests on one institutional fact: for this material class, the research, the pilot plant and the testing laboratory sit inside a single university, and have done so for three decades.

The State Key Laboratory of Advanced Papermaking and Paper-based Materials was approved by the Ministry of Science and Technology in January 2025, reorganizing the former State Key Laboratory of Pulp and Paper Engineering, which was built in 1989 and opened in 1996 as the first national key laboratory in China's light industry. It is jointly hosted by SCUT and China Banknote Printing and Minting Corporation, and works across plant-fiber chemistry, green pulping, high-performance paper-based materials and security fibers. Its discipline, Light Industry Technology and Engineering, holds an A+ rating in the national assessment. The laboratory occupies more than 21,000 square meters, holds instruments valued at about RMB 305 million, and its testing platform covers the whole chain from pulping to converting against ISO, TAPPI and GB methods.

Method development. The laboratory develops and validates test methods for fiber-based barrier materials, coating performance and repulpability, and operates them routinely rather than on request. This is the part of the work that cannot be subcontracted to a consultancy, because the method has to be written by people who run the test.

Pilot translation. The National Engineering Research Center for Paper and Pollution Control provides pilot and demonstration capacity between laboratory scale and commercial production. A method that cannot be executed at pilot scale is a method that will fail in a factory.

Testing and conformity. The testing platform covers barrier performance, grease resistance, heat seal, repulpability and food-contact safety to ISO, TAPPI and GB methods. For a conformity tool this matters twice over: the laboratory that writes the method has to be able to reproduce it, and an independent laboratory has to be able to repeat the result from the published template alone.

Industrial data. Cooperating enterprises supply data from commercial lines, which is the only way to test a method against the variability of real manufacturing, where fiber furnish, coat weight and machine speed never match the laboratory sample.

SCUT would meet the direct costs of the first phase from its own research resources. An indicative budget can be provided on request; it is not a request for UNIDO funds.

One disclosure belongs here. Several academic staff of the laboratory are founders or shareholders of companies active in this field, including the partner described in Section 5. We do not think this disqualifies the work, since the same is true of most applied materials research, but it has to be managed rather than asserted away. We therefore propose that the methodological review be chaired by an independent testing or certification body acceptable to IET/PST, and that no affiliated company take part in the review of the method it would itself use.

5. Industrial partners

A method written in a laboratory and never run on a production line is of limited use. The partners below supply the production data, the pilot lines and, most usefully, the objections that a working method has to

survive.

Guangzhou Shuhe Technology Co., Ltd. develops waterborne, bio-based barrier coatings intended to replace three incumbent technologies in food-contact packaging: polyethylene extrusion coating, acrylic dispersion coating, and fluorinated greaseproof treatment. Its products are applied to food packaging paper and to molded fiber, and include water-and-oil barrier coatings, an abrasion-resistant coating for molded fiber industrial packaging, a heat-seal coating for single-use food service items, and a water-based overprint varnish. The company was established in 2020 in Zengcheng, Guangzhou, by researchers from the laboratory, holds a plastic-free verification from SGS, and closed a Series A round in 2023.

Shuhe's relevance to this proposal is specific rather than general. The coating layer is where a plastic-free claim is most often contested: a fiber substrate contains no plastic by nature, but its barrier performance usually comes from a polymer, and whether that polymer counts as plastic depends on the chemistry, the coat weight and the definition being applied. A method that cannot settle that question at the coating layer will not survive contact with a European importer. Shuhe holds the formulation, coat weight and barrier data needed to define functional equivalence, and its existing third-party plastic-free verification gives the methodology a working precedent to build on instead of a new claim framework to invent.

The other partners cover the remaining links in the chain: Qingqing New Materials for coating and functional additives, BOIT for process-data modeling on pulp and paper lines, and Zhuzhou Times Huaxian for high-performance paper-based materials. Leading paper and packaging producers, including APP Sinar Mas, Nine Dragons Paper, Sun Paper and Yuto Packaging, are expected to contribute production-line data and to act as early users of the method. [Partner names to be used only after each company has confirmed in writing.]

Two rules apply to all of them. Production data enter the method in anonymized and aggregated form, and no proprietary formulation is published. Results are reported at the level of technology class rather than company, so that the mapping in Deliverable 2 cannot be read as a supplier ranking.

6. Deliverables and acceptance criteria

Three outputs are proposed for an eighteen-month first phase. Each carries a stated scope, a form of validation and an acceptance test, on the principle that a deliverable without an acceptance test cannot be reviewed.

Deliverable 1: Methodology for measuring plastic substitution, Draft V0.1

The methodology addresses the three questions in Section 2 and stops there. It is issued in three parts: a specification of the method, a calculation workbook that implements it, and a reporting template that a firm can complete without an LCA consultant.

(a) Biogenic carbon: the timing convention for uptake and release, the treatment of carbon stored in durable products, and the horizon over which delayed emissions are counted.

(b) Substitution boundary: definition of the displaced incumbent, a functional equivalence test covering barrier, grease resistance and heat seal, and the end-of-life scenario set, with default values for cases where primary data are unavailable.

(c) End-of-life and recyclability: a decision tree that maps a product and a market to the applicable test, so that a claim states which regime it satisfies rather than asserting that it is compliant.

Validation runs the draft against production data for at least three product families, namely molded fiber food containers, coated food-service paper and paper-based industrial packaging. A result counts as reproducible when an independent laboratory reaches the same figure, within a stated tolerance, from the published template alone.

Two testing or certification bodies, at least one of them nominated by IET/PST, review the draft. Their comments and our responses are published with V0.1, including the comments we do not accept.

Where the method touches European requirements the intention is alignment, not equivalence. The reporting template uses the parameter names and units of the Product Environmental Footprint method where they exist, so that a completed template can be converted into a PEF study rather than repeated. We make no claim that conformity with this method confers conformity with the PEF method, or with the PPWR.

Deliverable 2: Technology and industry mapping

The mapping is bounded in scope: five technology families, covering molded fiber, coated paper and board, fiber-based flexible laminates, bio-based coatings and fiber-reinforced composites; at least forty technology entries; and coverage of at least twenty economies. Each entry records the performance envelope (barrier, grease resistance, heat resistance, maximum service temperature), an indicative cost range, the product categories it can serve, its maturity level, and the substitution ratio actually achieved in commercial use.

Sources are patents, customs data including HS 4823.70 for molded pulp articles and HS 4811 for coated paper and board, company disclosures, and the laboratory's own industry contacts. Maturity is assigned against a published rubric, and the rubric is itself part of the deliverable, so that a reader who disagrees with a rating can see what produced it.

The mapping is not a supplier catalog and does not rank vendors. It also carries the substitution-suitability criteria, a short set of tests that tells a policy maker which product categories a fiber substitute can realistically serve and which it cannot. This is the part most likely to be contested, because it will conclude that some substitutions are not sensible. We would rather say so in the document than have it said about the document.

Deliverable 3: South–South technical dialogue

One technical forum, convened under UNIDO's auspices, of sixty to eighty participants drawn from policy, industry and research in producing countries. It is a working meeting and not a conference: the agenda is the review of Draft V0.1 by the firms that would have to use it, and the output is a written set of amendments that goes back into the revision.

Alongside it, a two-day training module on the method, piloted in one region agreed with PST. Southeast Asia or East Africa would be our suggestion, where fiber raw material and export exposure coincide, but the choice is properly PST's.

A caution is due on this deliverable. A single forum is a weak instrument for building a network. It is proposed because it is what can be delivered credibly in a first phase, and it should be judged as a test of demand

rather than as the founding of an institution.

7. Governance and legal arrangements

The form would be a Memorandum of Understanding between UNIDO and SCUT, with SCUT as legal host. That is the standard vehicle for this kind of arrangement and we propose no deviation from it.

A joint steering committee, co-chaired by IET/PST and SCUT, would approve an annual work plan and receive the deliverables twice a year. A secretariat at the laboratory would carry out coordination, data handling and the review process, staffed by two people, one of them full-time.

The wider network of participating institutions would exist as a cooperation arrangement without legal personality. It would not contract, hold funds or employ staff in its own name. No registration as a legal entity is proposed at this stage. Any future decision to seek legal personality would be taken by the members, in the appropriate jurisdiction and subject to the required approvals, and lies outside the scope of this note.

Nor would the arrangement use UNIDO's name or emblem in its own title. We propose instead that it be described as a cooperation network hosted by SCUT and associated with the work of IET/PST, with UNIDO's involvement recorded in the text of the Memorandum rather than carried in the network's name. This follows UNIDO's rules on the use of its name, and it is also the more durable arrangement: a network that borrows its authority has less reason to build any of its own.

Intellectual property would be settled before signature rather than after. The specification and the template would be released under an open license, CC BY 4.0 for the text and a permissive license for the workbook, with authorship credited to SCUT. Background intellectual property of the industrial partners is unaffected.

Entity	Role	Contribution
UNIDO (IET/PST)	Technical guidance, convening, dissemination	Review of scope; nomination of one independent reviewer; convening of the South–South dialogue; dissemination through country programs already under delivery
SCUT	Legal host and operational anchor	Laboratory and pilot facilities; method development and testing; secretariat; direct costs of the first phase
Industrial partners, including Shuhe Technology	Production data, pilot validation, user feedback	Anonymized data from commercial lines; pilot runs; comment on the draft from the position of a user
Testing and certification bodies	Independent review, conformity interface	Review of Draft V0.1; assessment of the template against audit practice; advice on recognition routes
Producing-country institutions	Demand definition and uptake	Participation in the dialogue; pilot application in at least two countries, to be identified with PST

If UNIDO prefers a lighter start, the methodology review and the dialogue can proceed without a Memorandum, on the basis of an exchange of letters limited to those two activities. We would prefer the fuller arrangement, but a lighter one that begins this year is worth more than a fuller one that begins next year.

8. Work plan

The plan assumes a start within one month of technical alignment. Month 1 is counted from the first meeting with the focal point, not from the date of this note.

Period	Milestone
Months 1–3	Focal point designated; scope confirmed; data agreements signed; review panel constituted
Months 4–6	Method framework and data protocol issued for comment; mapping rubric fixed; first collection covers 10 economies
Months 7–9	Draft V0.1 issued; first full validation run on molded fiber; workbook released in beta
Months 10–12	Independent review of V0.1; mapping V1 published; training module drafted
Months 13–18	Method V1.0 issued with review responses; pilot training in one region; South–South dialogue held; phase-two proposal submitted

Two assumptions are worth stating. The validation data have to be available by month 7 or the schedule moves; the partners have indicated that they can meet this, but it is the item we would watch. And the external review in months 10 to 12 depends on reviewers being available in that window, which is why we ask for one of them to be nominated early.

9. Risks

Four risks are material enough to set out.

Legitimacy. A method developed by one university, in one country, with industrial partners that make the products it measures, will be suspected of convenience. The mitigation is structural rather than rhetorical: an independent chair for the review, a sixty-day public comment period on Draft V0.1, publication of all comments with responses, and a review panel on which no affiliated company sits.

Data. Firms will not release production data that reveals cost structure or formulation. The method is therefore designed to run on a minimum data set that a firm can supply without disclosure, aggregated at technology-class level, under a data handling protocol agreed before collection begins.

Recognition. Neither SCUT nor UNIDO can deliver recognition of the method by the European Commission or by ISO. What can be delivered is a template that converts, a review record that stands up to scrutiny, and a route into the relevant technical committees. Anything stronger would be a promise we could not keep.

Continuity. Arrangements of this kind usually lapse when the founding professor retires or the first grant ends. The secretariat is therefore staffed rather than honorary, and the second phase is designed to be financed by service income, from firms and from country projects that use the method, rather than by a second grant.

The largest risk, though, is scope. Each of the three deliverables could absorb eighteen months on its own. The boundary is fixed in Section 6, and anything outside it is deferred in writing to a second phase rather than absorbed quietly into the first.

10. Results framework

As a public good, the initiative should be judged on use rather than on publication. The first phase would be measured against the following indicators, with the baseline as of the start of the work.

Indicator	Target at month 18
Independent laboratories reproducing a published result, in different regions	2
Firms completing the template without external consultancy	10, of which at least 5 in low- or lower-middle-income countries
Economies covered in the published mapping	20
Testing or certification bodies applying the method in their own assessment	1
Participants in the dialogue from producing countries	40

Two of these are worth more than the others. If no independent laboratory can reproduce a result from the published template, the method has not been specified. If no firm completes the template without hiring a consultant, the problem in Section 2 has not been solved, only restated.

11. What is requested

Three things are requested, in order of importance.

A technical focal point in IET/PST. We estimate the review load at five to eight person-days over eighteen months, concentrated in months 1 to 3 and months 10 to 12. Staff time, rather than money, is the scarce resource in arrangements of this kind, and the request has been sized accordingly.

Advice on the form of engagement: a Memorandum with SCUT, participation in the methodology review, or a convening role in the dialogue. We have a preference for the first, but the choice should follow the technical assessment rather than precede it.

An indication of fit with PST's work program for 2027, so that we can sequence our own resourcing. If the topic is better placed elsewhere in the Directorate, we would rather be told so now.

[To be confirmed by SCUT before transmission:] SCUT's leadership has authorized this proposal and would meet the costs of the first phase. The President would welcome the opportunity to meet the Directorate in Vienna at a time convenient to UNIDO. No date is proposed here, since the timing should follow the technical review rather than anticipate it.

A detailed work plan, a data handling protocol and an indicative budget can be provided within four weeks of the first meeting. Correspondence on this note may be addressed to the Office of International Cooperation, South China University of Technology, [contact name, email], with technical questions to the State Key Laboratory [contact name, email].

Annex ｜ 中文摘要（供校内审议）

一、定位。本文件是向联合国工业发展组织（UNIDO）SDG创新与经济转型司（IET）公平生产、可持续标准与贸易处（PST）提出的技术合作建议。由华南理工大学承担方法与数据工作，在18个月内交付三项可直接嵌入PST现有服务包的公共产品；不要求UNIDO出资，不新设法人机构，不需要UNIDO增加人员。

二、问题。植物纤维替代塑料的技术已进入商业化，具有巨大的环境与气候变化效益，但全球现状是缺少该可持续材料替代塑料的产业共识。争议集中在三处：一是生物碳的核算时点，ISO 14067只给原则，未定时点惯例，同一产品的两项研究可因此相差甚大；二是替代边界的设定，被替代物、末端处置情景与功能单位均由分析者选取而非实测，末端情景一项即可使结果相差一倍以上；三是可回收与可降解声明的适用规则，EN 13432、ISO 14855/17556、各国回用纤维方案以及欧盟《包装和包装废弃物法规》(EU) 2025/40（2026年8月12日起适用）各不相同，一个产品可能满足其一而不满足其二。后果主要落在发展中国家出口企业：一份完整的第三方碳足迹研究需数万欧元，中型企业无法按产品线逐一承担，这实际上构成了贸易壁垒。

三、交付物。（一）《植物纤维材料替代塑料核算方法（V0.1）》，含方法规范、计算工作簿和可审计报告模板，覆盖上述三个问题；以三家以上产业伙伴的真实生产数据验证，由两家检测认证机构（其中一家由PST提名）评审，评审意见与本方答复随V0.1一并公开。（二）《植物纤维材料技术—产业图谱》，限定五类技术、40个以上技术条目、20个以上经济体，标注性能边界、成本区间、成熟度与实测替代比，并附替代适用性判据。（三）在UNIDO主持下的南南技术对话，规模60至80人，为工作性会议而非一般研讨会，同期交付两天期培训模块，试点区域与PST商定。

四、我方条件。先进造纸与纸基材料全国重点实验室于2025年1月经科技部批准，由原制浆造纸工程国家重点实验室重组，华南理工大学与中国印钞造币集团有限公司共同建设；原实验室1989年立项、1996年开放运行，是我国轻工领域首个国家重点实验室。实验室面积逾2.1万平方米，仪器设备总值约3.05亿元，检测平台覆盖制浆、抄造、涂布、检测全流程，按ISO、TAPPI、GB方法运行；造纸与污染控制国家工程研究中心承担中试验证。依托“轻工技术与工程”A+学科。

五、产业伙伴。拟合作伙伴包括：全球头部造纸企业金光纸业（APP）、玖龙纸业、太阳纸业；头部包装企业裕同科技；SCUT孵化企业广州树合科技有限公司（生物基阻隔涂层）；以及Qingqing New Materials（涂层与功能助剂）、BOIT（制浆造纸过程数据建模）、株洲时代华先（高性能纸基材料）。【企业中文全称及是否同意具名，须在发出前逐一书面确认】

六、需说明的一点。实验室内有教师是相关企业的创始人或股东，包括广州树合科技有限公司。我们并不认为这构成回避理由，但须有制度安排，故建议方法学评审由PST认可的独立检测认证机构主持，关联企业不参与与其自身产品相关条目的评审。

七、治理与法律形式。以华南理工大学为法律主体与UNIDO签署谅解备忘录；设联合指导委员会，由PST与华南理工大学共同主持；秘书处设于实验室，配备专职人员2名，其中1人全职。合作网络初期为无法人资格的合作安排，不对外签约、不设账户、不单独用人。按现行规定，此类网络不能在我国境内登记为国际组织，本件不提出登记诉求；今后如需取得法人资格，由成员另行决定并另行报批。网络名称不使用UNIDO名称与标识，对外表述为“由华南理工大学主办、与PST工作相关联的合作网络”，UNIDO的参与在备忘录正文中体现。

八、请求。请PST指定技术对接人，评估方法学范围与交付物，并就合作形式提出意见；预计评审工作量五至八人日，集中在第1至3月与第10至12月。【以下两项须经学校正式同意后方可保留】华南理工大学已同意承担第一阶段直接费用；校长愿在UNIDO方便时赴维也纳拜会，具体时间服从技术评审进度，本件不预设日期。